CS 6460 · Utah Valley University

Causally-Constrained Counterfactual Explanations

for GC-MS Retention Time Prediction

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Agenda

1. **Motivation** — why causality in XAI?
2. **Pipeline** — four-step system
3. **Causal graph** — the DAG
4. **Results I** — violation analysis
5. **Results II** — metric comparison
6. **Results III** — proximity trade-off
7. **Results IV** — RT robustness
8. **Case studies**
9. **Discussion**
10. **Conclusion & future work**

At a glance: 16 molecules · 80 counterfactuals · **95% causal pass rate** → **100%** after constraints · proximity cost: $r{=}{-}0.14$, $p{=}0.603$ (n.s.)

The problem with standard counterfactuals

- DiCE optimises for *statistical* validity & diversity
- No concept of chemical plausibility
- Can suggest: *"increase surface area while decreasing MW"* — physically impossible
- Downstream descriptors change without upstream causes moving

Core problem

Model-valid $\neq$ chemically valid.
Useless — and misleading — for a practitioner.

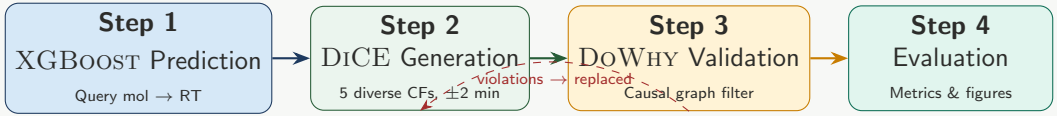
Our solution

Causal constraint layer

- Hand-crafted DAG from chemistry domain knowledge
- DoWhy validates every counterfactual against causal ordering
- Violations flagged & replaced
- Result: physically interpretable explanations

Grounded in Pearl's structural causal model [3, 6]

Pipeline Overview | Four-step causally-constrained system



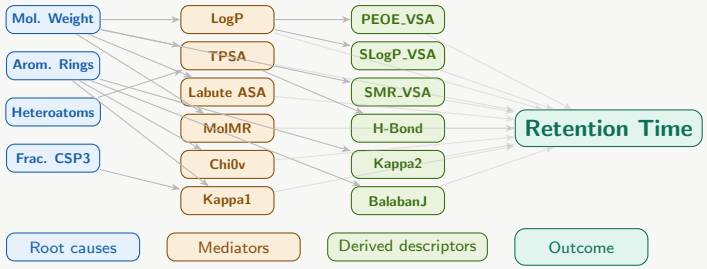
Foundation model (CS 6610 capstone)

- XGBoost on 2,000 molecules, 1,900 measurements
- 37 RDKit features across 15 GC columns
- R^2=0.9714, MAE = 0.74 min

New contributions (CS 6460)

- DiCE counterfactual generation
- DoWhy causal graph & validation
- Evaluation framework (5 experiments)

Causal Graph | DAG encoding analytical chemistry domain knowledge



Directed edges enforce causal ordering. DoWhy checks: if a child descriptor changes, did its parent change consistently?

95%
causal pass rate
(76/80 CFs)

100%
after enforcement
(16/16 molecules)

4 violations
all in *one* molecule:
`CCN(CC)CC`
(triethylamine)

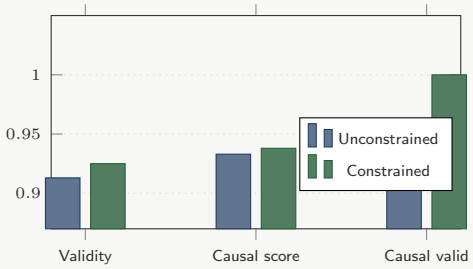
Key findings

- **76/80** CFs causally valid *without* intervention
- **15/16** molecules required *zero* filter intervention
- All 4 violations in **triethylamine**:
  branched tertiary amine creates complex descriptor dependencies that DiCE's unconstrained search violated
- Causal score distribution shifts **right** after filtering
- Causal valid rate: $0.950 \rightarrow \mathbf{1.000}$

Why triethylamine? Its branched tertiary amine creates unusual correlations between shape indices (Kappa1, Kappa2) and connectivity descriptors that DiCE violates when optimising for RT proximity alone.

Results II | Unconstrained vs. constrained pipeline metrics

Metric	Uncon.	Con.	Δ
Validity	0.913	**0.925**	$+0.012$
Causal score	0.933	**0.938**	$+0.005$
Causal valid rate	0.950	**1.000**	$+0.050$



Headline result

The DoWhy layer is **strictly additive**.

There is **no metric** where the constrained pipeline performs worse.

Per-molecule validity vs. causal score

- 2 outliers with validity < 1.0: acetophenone & a polycyclic aromatic
- Both have causal scores ≈ 0.948 — same as the high-performing cluster
- Failures $\Rightarrow$ *molecular complexity*, not constraint interference

The anticipated concern

- Constraints might *force* CFs further from query
- Higher L1 distance ⇒ less actionable
- Well-documented in XAI literature [4, 5, 11]

Correlation analysis

Condition	r	p
Unconstrained	$+0.18$	0.493
Constrained	-0.14	0.603

Neither result is statistically significant.

Key finding — refutes the criticism

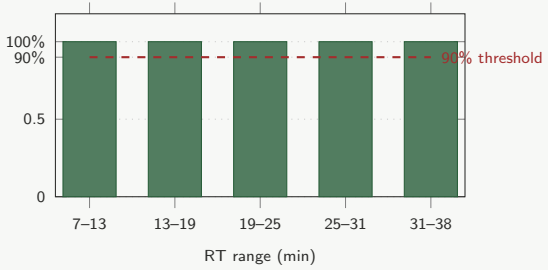
No significant relationship between proximity and causal plausibility in either condition.

Causal constraints do **not** systematically degrade explanation actionability.

15/16 molecules see essentially **no proximity shift** after constraints.
Only triethylamine increases:
L1 distance $\approx 40 \rightarrow \approx 90$.

Test design

- 16 molecules binned into 5 equal RT ranges (7.1–37.5 min, ≈6 min each)
- Pass threshold: 90% causal valid rate



All five bins at 100%

No RT-range bias detected. The causal graph holds across the full chromatographic range (7–38 min).

Why this matters

- GC-MS retention is physically different at each extreme:
  - *Early eluters:* volatility-dominated
  - *Late eluters:* polarity / boiling point
- Uniform performance ⇒ DAG captures **genuine physicochemical structure**, not incidental correlations

Feature Change Analysis | Which descriptors does DiCE manipulate most?

Feature	CF1	CF2	CF3	CF4	CF5
labuteASA	.85	.78	.91	.70	.82
peoe_vsa2	.60	.72	.55	.80	.45
slogp_vsa2	.75	.68	.82	.60	.88
logP	.40	.55	.62	.48	.70
kappa2	.10	.12	.08	.15	.95
molmr	.10	.12	.08	.15	.10
balabanj	.12	.10	.13	.09	.11
num_arom.	.20	.18	.22	.19	.21

Green = active; gray = held stable; CF5* is structurally distinct

Active levers (high variation)

- labuteASA, peoe_vsa2, slogp_vsa2 — surface area descriptors
- logP — lipophilicity
  Chemically meaningful RT drivers [9]

Held stable by causal constraints

- molmr, balabanj
- num_aromatic_rings near-constant
- Consistent with DAG: downstream descriptors cannot change without upstream causes

CF 5 is structurally distinct — evidence of genuine diversity in the constrained set.

Experiment 3 — Case Studies | Best, mid, and worst-case performance

↑ Best case

`CCNCC`
Diethylamine

- Perfect validity & causal rate
- Zero constraint intervention
- Unconstrained = constrained
- Proximity ≈ 35, diversity ≈ 4

= Mid case

`CCCCCCCCCC`
Decane

- Stable performance
- High diversity ≈ 43
- Many paths to reduce RT for a C_{10} alkane
- Both pipelines identical

↓ Worst case

`CCN(CC)CC`
Triethylamine

- Unconstrained rate ≈ 0
- **Constrained rate** $= 1.0$
- Proximity: $\approx 40 \rightarrow \approx 90$
- Diversity $\rightarrow$ near zero: single valid causal path

Key insight: A causally *invalid* explanation close to the query is **worse** than a causally valid one further away. Triethylamine's unconstrained CFs suggest chemically impossible modifications.

Discussion

What the results show

- DoWhy is **strictly beneficial** — no metric regresses
- Proximity and causal plausibility are **statistically independent** ($r{=}-0.14$, $p{=}0.603$)
- Directly **refutes** the criticism that constraints degrade actionability [4, 11]
- Uniform performance 7–38 min confirms the DAG captures **genuine** physicochemical structure, not incidental correlations

The triethylamine lesson

- Branched tertiary amine ⇒ most complex descriptor dependencies of any test molecule
- Only case where unconstrained DiCE fails to find causally valid CFs
- Causal layer sacrifices proximity for plausibility — the correct trade-off

MW, LogP, surface area — the primary nodes in the DAG — confirmed as dominant GC-MS RT drivers by Liu et al. [10].

Conclusion & Future Work

Key contributions

1. Hand-crafted **causal DAG** encoding 37 RDKIT descriptors from domain knowledge
2. **Four-step pipeline:** XGBOOST → DICE → DOWHY → evaluation
3. **95% causal pass rate → 100%** no systematic proximity cost (r=−0.14, p=0.603)
4. **No RT-range bias** across 7–38 min
5. Chemically interpretable worst-case analysis (triethylamine)

Future work

- Expand to the full **RepoRT dataset** [12]
- Extend causal graph to cover **all 37 features**
- Explore **learned causal structures** from observational data

Causal constraints are strictly free: validity ↑, plausibility ↑, proximity unchanged.

Thank you

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Questions?

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